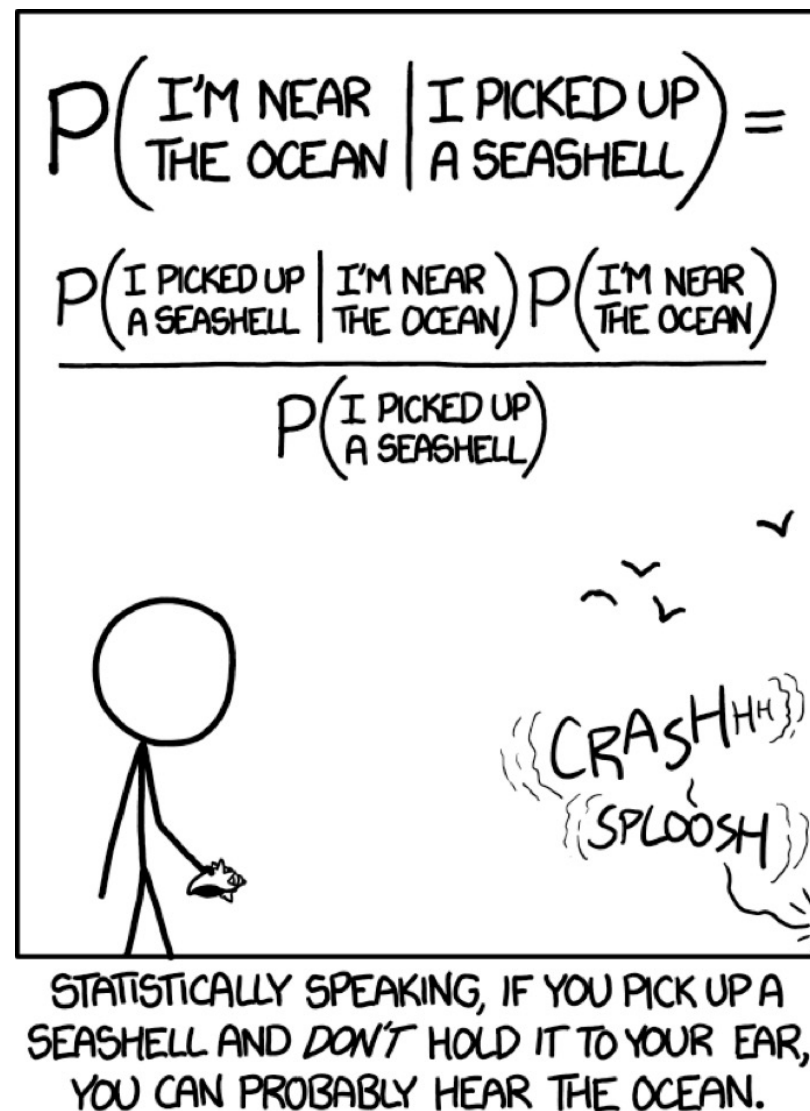


CS 237: PROBABILITY IN COMPUTING

INSTRUCTORS: TIAGO JANUARIO, NATHAN KLEIN

BOSTON
UNIVERSITY



<https://xkcd.com/1236/>

LECTURE 10

Last time

- Functions of random variables
- Conditional Probability
- Chain rule

Today

- Quiz 2
- Law of Total Probability
- Bayes Rule
- Independence

RECAP

- ▶ We learned what conditional probability is.

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$$

- ▶ We showed that $\mathbb{P}(\cdot \mid B)$ is a probability function (satisfies the axioms)
- ▶ We learned the **chain rule** (which is really just the definition of conditional probability, but is helpful anyway)

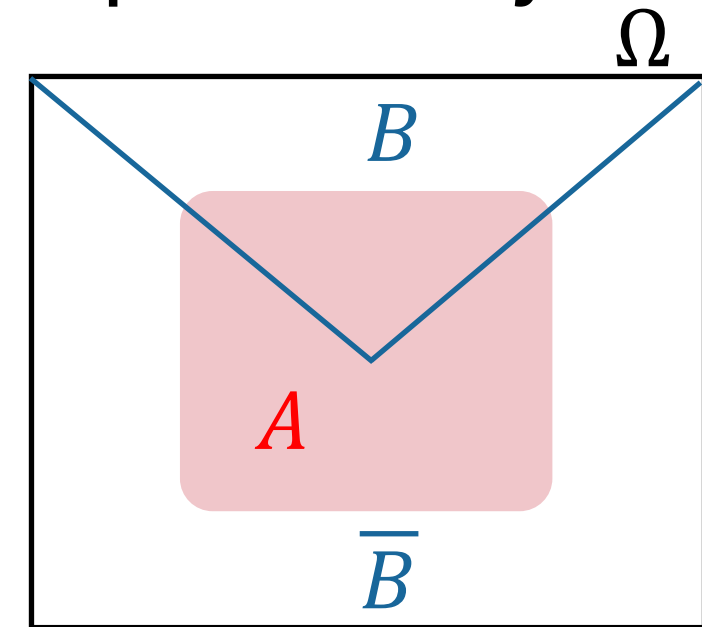
$$\mathbb{P}[A \cap B] = \mathbb{P}[A \mid B] \mathbb{P}[B]$$

- ▶ Next up: **law of total probability** and **Bayes Theorem**

LAW OF TOTAL PROBABILITY

- ▶ We can use conditional probability to break down probability calculations

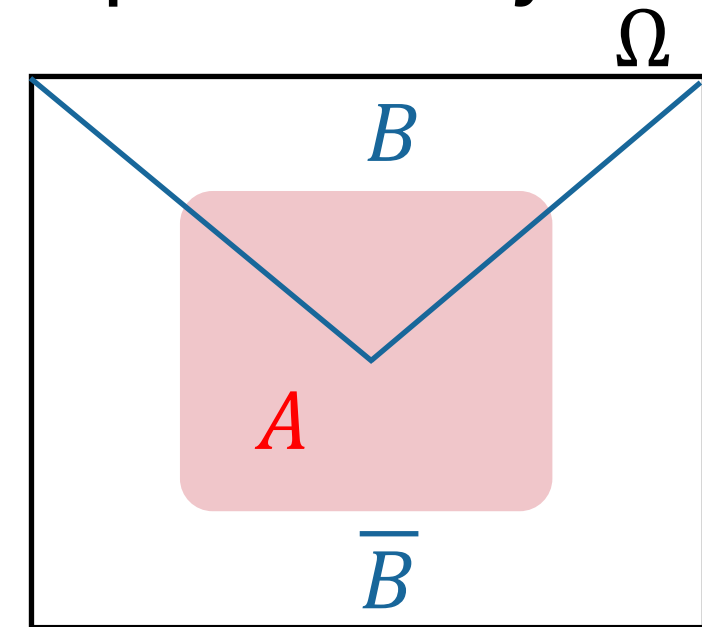
$$\mathbb{P}(A) = \mathbb{P}(A \mid B) \cdot \mathbb{P}(B) + \mathbb{P}(A \mid \bar{B}) \cdot \mathbb{P}(\bar{B})$$



LAW OF TOTAL PROBABILITY

- ▶ We can use conditional probability to break down probability calculations

$$\mathbb{P}(A) = \mathbb{P}(A \mid B) \cdot \mathbb{P}(B) + \mathbb{P}(A \mid \bar{B}) \cdot \mathbb{P}(\bar{B})$$



We have 2 coins: one normal, one double-headed

We pick a coin uniformly at random and toss it

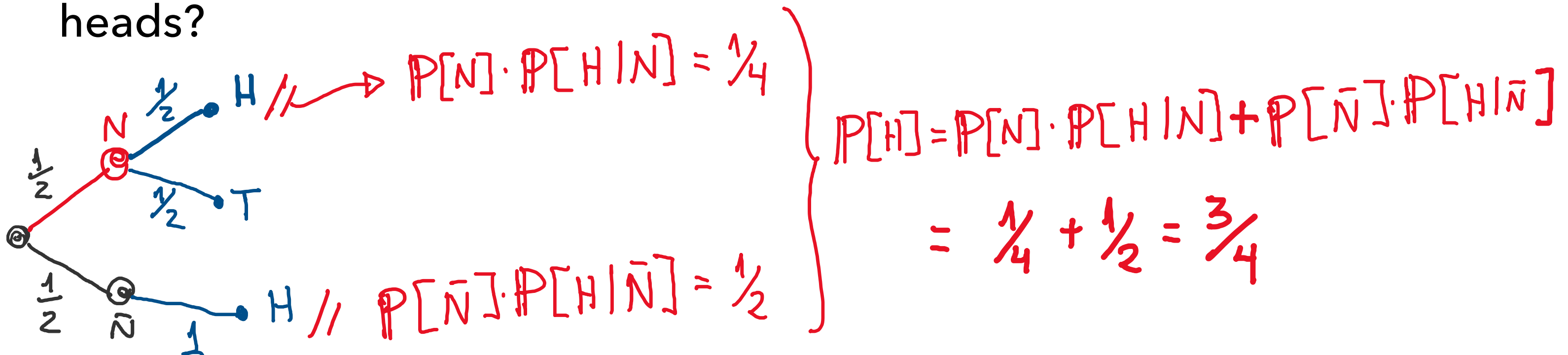
What is the probability that we get a heads?

LAW OF TOTAL PROBABILITY

- ▶ We can use conditional probability to break down probability calculations:

$$\mathbb{P}(A) = \mathbb{P}(A \mid B) \cdot \mathbb{P}(B) + \mathbb{P}(A \mid \bar{B}) \cdot \mathbb{P}(\bar{B})$$

- ▶ We have 2 coins: one normal, one double-headed. We pick a coin uniformly at random and toss it. What is the probability that we get a heads?



LAW OF TOTAL PROBABILITY: PROOF

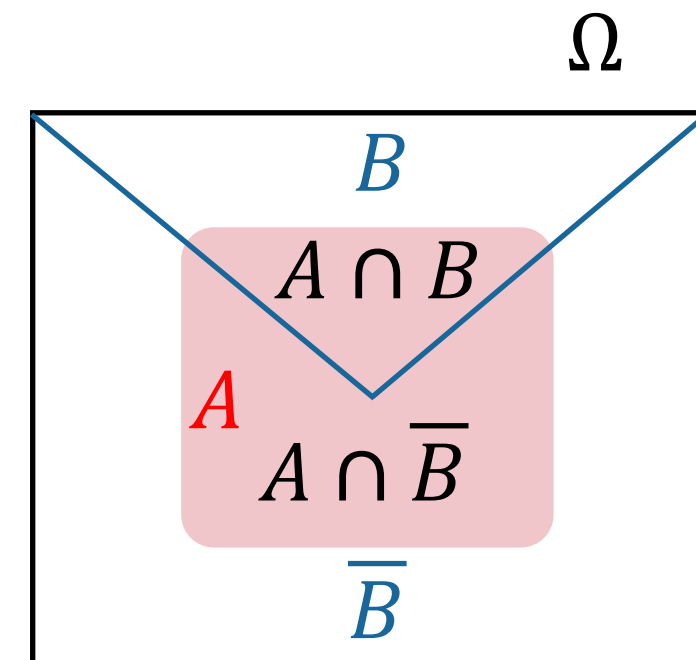
- Law of total probability:

$$\mathbb{P}(A) = \mathbb{P}(A | B) \cdot \mathbb{P}(B) + \mathbb{P}(A | \bar{B}) \cdot \mathbb{P}(\bar{B})$$

$$P[A] = P[(A \cap B) \cup (A \cap \bar{B})] \quad (\text{Property of sets})$$

$$= P[A \cap B] + P[A \cap \bar{B}] \quad (\text{additivity})$$

$$= P(A|B) \cdot P(B) + P(A|\bar{B}) \cdot P(\bar{B}) \quad (\text{chain rule})$$



LAW OF TOTAL PROBABILITY

- Can generalize to larger partitions of the sample space:

$$\mathbb{P}(A) = \mathbb{P}(A \mid B_1) \cdot \mathbb{P}(B_1) + \mathbb{P}(A \mid B_2) \cdot \mathbb{P}(B_2) + \mathbb{P}(A \mid B_3) \cdot \mathbb{P}(B_3)$$

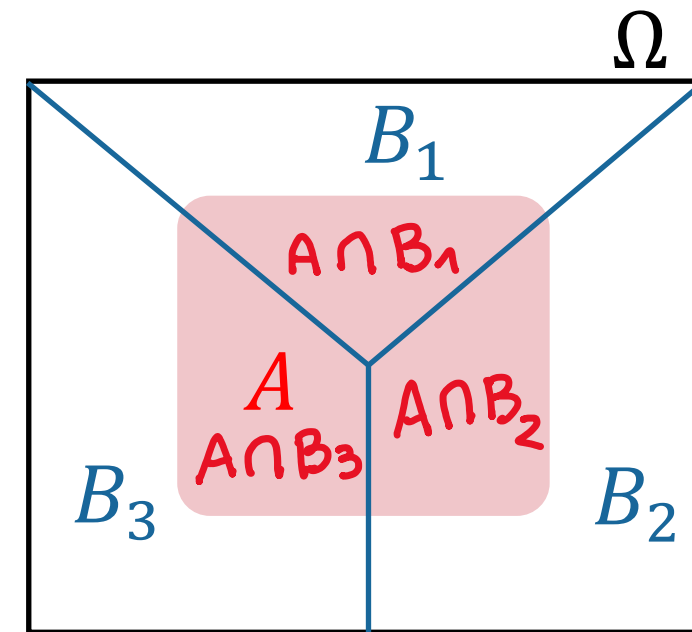
You already know this from tree diagrams!

General form:

Let B_i be partitions of Ω .

Then $\forall A \subseteq \Omega$:

$$P_r(A) = \sum_{i=1}^k P_r(A \mid B_i) \cdot P_r(B_i)$$



LAW OF TOTAL PROBABILITY

- Can generalize to larger partitions of the sample space:

$$\mathbb{P}(A) = \mathbb{P}(A \mid B_1) \cdot \mathbb{P}(B_1) + \mathbb{P}(A \mid B_2) \cdot \mathbb{P}(B_2) + \mathbb{P}(A \mid B_3) \cdot \mathbb{P}(B_3) \quad \sim$$

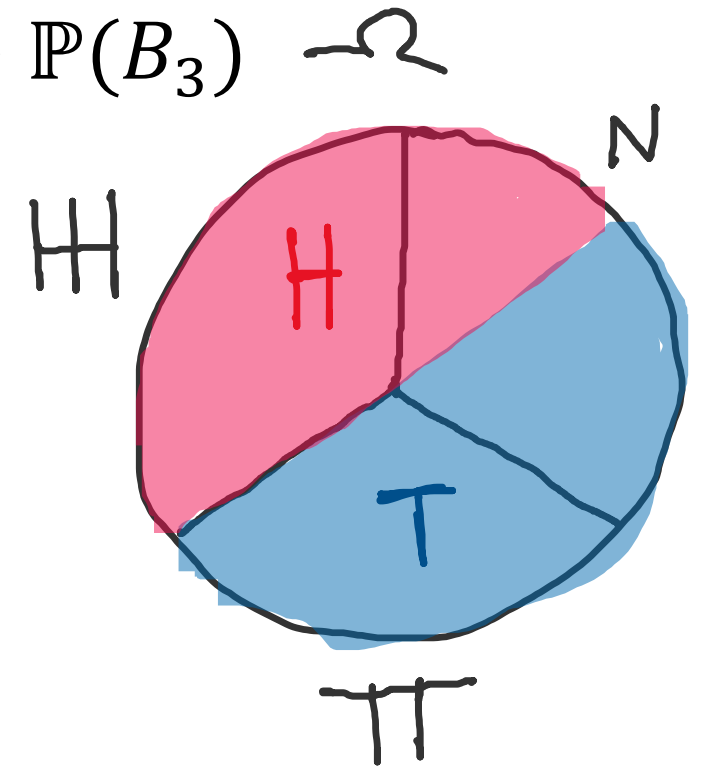
3 coins: normal, double-headed, double-tailed

We pick a coin uniformly at random and toss it

What is the probability we get heads?

$$P(H) = P(H|N) \cdot P(N) + P(H|TT) \cdot P(TT) + P(H|HH) \cdot P(HH)$$

$$= \frac{1}{2} \cdot \frac{1}{3} + 0 \cdot \frac{1}{3} + 1 \cdot \frac{1}{3} = \frac{1}{2}$$



ASK THE AUDIENCE

- ▶ First, we toss a coin
- ▶ If the coin is heads, we roll a fair die and take the result
- ▶ If the coin is tails, we roll two fair dice and take the sum
- ▶ What is the probability of obtaining a 2?

$$(A) \frac{7}{72}$$

$$(B) \frac{1}{18}$$

$$(C) \frac{7}{36}$$

(D) None of the above

SOLUTION

- First, we toss a coin. If the coin is heads, we roll a fair die and take the result. Otherwise, we roll two fair dice and take the sum. What is the probability of obtaining a 2?

Let S = sum of the rolls

$$\Pr(S=2) = \Pr(S=2|H) \cdot \Pr(H) + \Pr(S=2|T) \cdot \Pr(T) \quad \left(\begin{array}{l} \text{law of total} \\ \text{probability} \end{array} \right)$$

$$= \frac{1}{6} \cdot \frac{1}{2} + \frac{1}{36} \cdot \frac{1}{2}$$

$$= \frac{1}{12} + \frac{1}{72} = \frac{6}{72} + \frac{1}{72} = \frac{7}{72}$$

ASK THE AUDIENCE

- ▶ We have two boxes with colored balls:
 - ▶ Box 1 has 2 red balls and 3 black balls
 - ▶ Box 2 has 8 red balls and 12 black balls
- ▶ We pick a ball uniformly at random from Box 1, and place it in Box 2
- ▶ We then pick a ball uniformly at random from Box 2
- ▶ What is the probability that the ball from Box 2 is red?

(A) $\frac{8}{21}$ (B) $\frac{9}{21}$ (C) $\frac{2}{5}$ (D) $\frac{4}{5}$ (E) None of the above

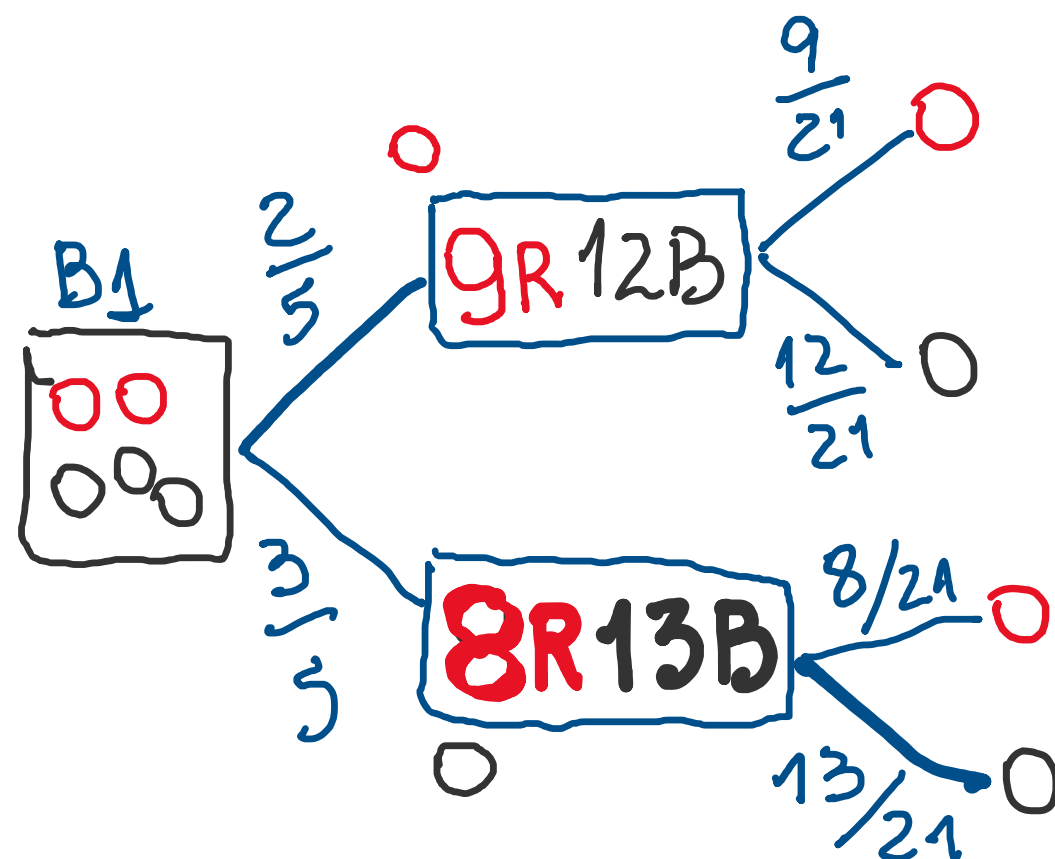
ASK THE AUDIENCE

Box 1: 2 red, 3 black

Box 2: 8 red, 12 black

- ▶ We pick a ball uniformly at random from Box 1, and place it in Box 2. We then pick a ball uniformly at random from Box 2. What is the probability that the ball from Box 2 is red?

C_{Red} = choose the red ball from Box 1



$$P(R \cap C_{\text{Red}})$$

$$P(R \cap \bar{C}_{\text{Red}})$$

$$P(R) = P(R|C_R) \cdot P(C_R) + P(R|\bar{C}_R) \cdot P(\bar{C}_R)$$

$$= \frac{3}{5} \cdot \frac{9}{21} \cdot \frac{2}{5} + \frac{8}{21} \cdot \frac{3}{5} = \frac{14}{35} = \frac{2}{5}$$

MEDICAL TESTING: BAYES THEOREM

- ▶ 0.8% of women ages 40-50 have breast cancer
- ▶ The mammogram test is the main screening tool
- ▶ The mammogram test is fairly accurate:
 - ▶ False negative rate is 10%: if a woman **has** breast cancer, the test is **negative** 10% of the time $P_r(P|C) = 0.9$
 - ▶ False positive rate is 7%: if a woman **does not have** breast cancer, the test is **positive** 7% of the time $P_r(P|\bar{C}) = 0.07$
- ▶ Suppose a woman takes the test and the test is positive
- ▶ What is the probability that she actually has breast cancer?

ASK THE AUDIENCE

- ▶ 0.8% of women ages 40-50 have breast cancer
- ▶ Mammogram test: false negative rate 10%, false positive rate 7%
- ▶ Suppose a woman between the ages of 40 and 50 is tested.

$\mathbb{P}(\text{cancer} \mid \text{test is positive})$ is

(A) $\leq 10\%$ (B) $\geq 90\%$ (C) 50% to 80%

ASK THE AUDIENCE

- ▶ 0.8% of women ages 40-50 have breast cancer
- ▶ Mammogram test: false negative rate 10%, false positive rate 7%
- ▶ Suppose a woman between age 40 and 50 is tested.

$\mathbb{P}(\text{cancer} \mid \text{test is positive})$ is

(A) $\leq 10\%$ (B) $\geq 90\%$ (C) 50% to 80%

Researchers asked American doctors:

95 out of 100 estimated the probability to be around 75%

ASK THE AUDIENCE

- ▶ 0.8% of women ages 40-50 have breast cancer
- ▶ Mammogram test: false negative rate 10%, false positive rate 7%
- ▶ Suppose a woman between age 40 and 50 is tested.

$\mathbb{P}(\text{cancer} \mid \text{test is positive})$ is

(A) $\leq 10\%$ (B) $\geq 90\%$ (C) 50% to 80%

Researchers asked 24 German doctors:

8 said (A), 8 said (B), 8 said (C)

SOLUTION

Events:

C = cancer , P = positive result

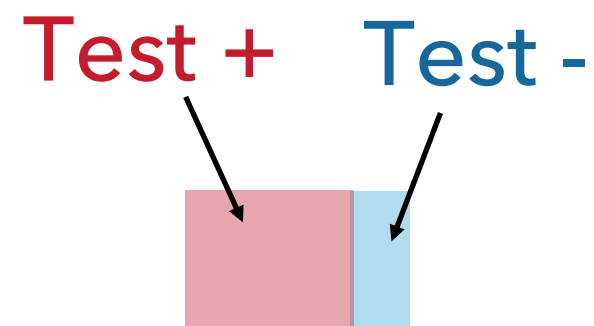
$$P_r(C|P) = \frac{P_r(C \cap P)}{P_r(P)} \quad (\text{definition of conditional probability})$$

$$P_r(C|P) = \frac{P_r(P|C) \cdot P_r(C)}{P_r(P|C) \cdot P_r(C) + P_r(P|\bar{C}) \cdot P_r(\bar{C})} \quad \begin{array}{l} (\text{chain rule}) \\ (\text{law of total probability}) \end{array}$$

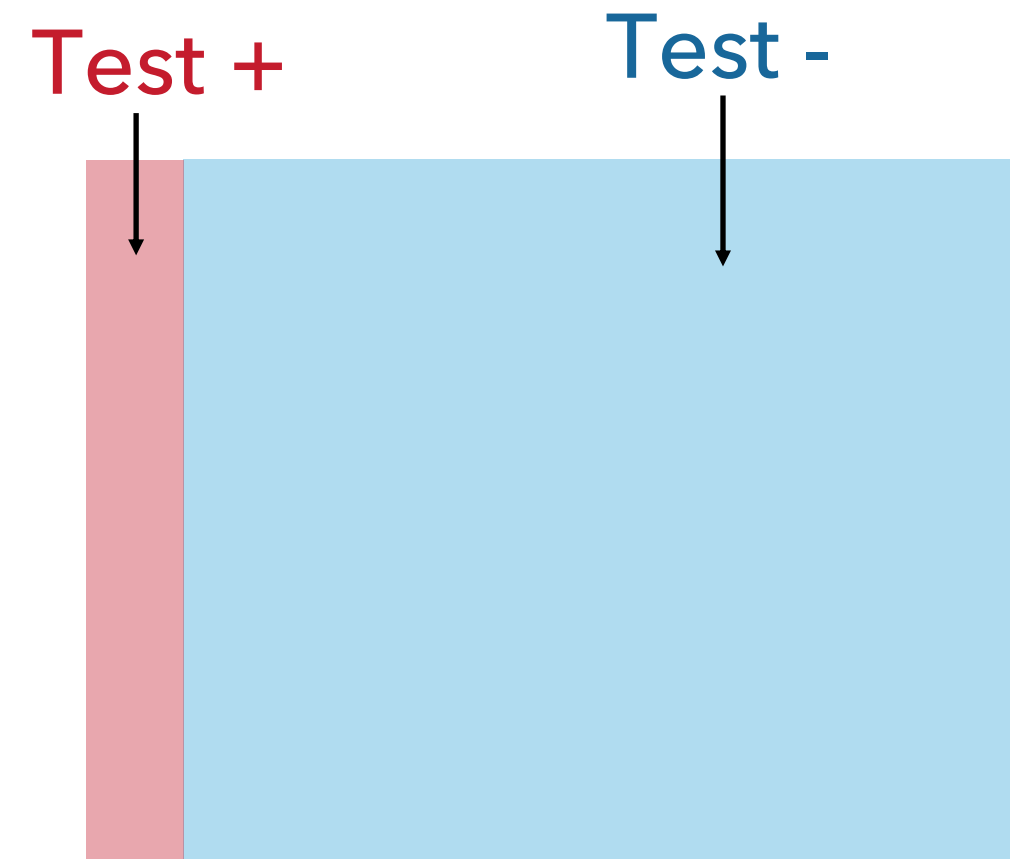
$$P_r(C|P) = \frac{0,9 \cdot 0,008}{0,9 \cdot 0,008 + 0,07 \cdot 0,992} \approx 0,0939 \approx 9,39\%$$

MEDICAL TESTING

- Most positive tests come from women **without** cancer



Have cancer
0.8%

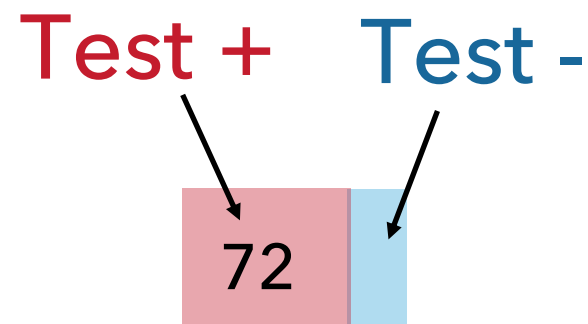


Do not have cancer
99.2%

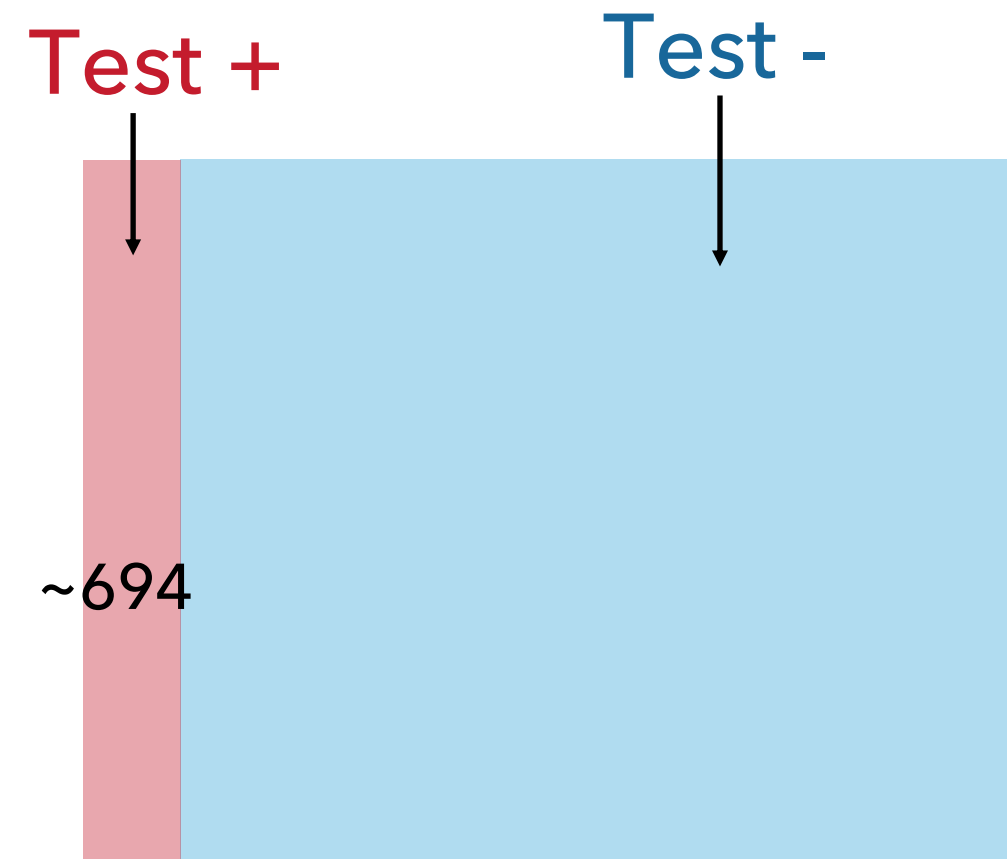
MEDICAL TESTING

$$P_r(C|P) = \frac{72}{72+694} \approx 9.4\%$$

- ▶ If the number of women is 10,000, we expect:



Have cancer
80



Do not have cancer
9920

MEDICAL TESTING

- ▶ **Experiment:** administer a mammogram test to a woman selected uniformly at random among women ages 40-50
- ▶ C = "woman has cancer"
- ▶ P = "test is positive"
- ▶ What we know:

$$\mathbb{P}(C) = 0.008$$

$$\mathbb{P}(P \mid C) = 0.9$$

$$\mathbb{P}(P \mid \bar{C}) = 0.07$$

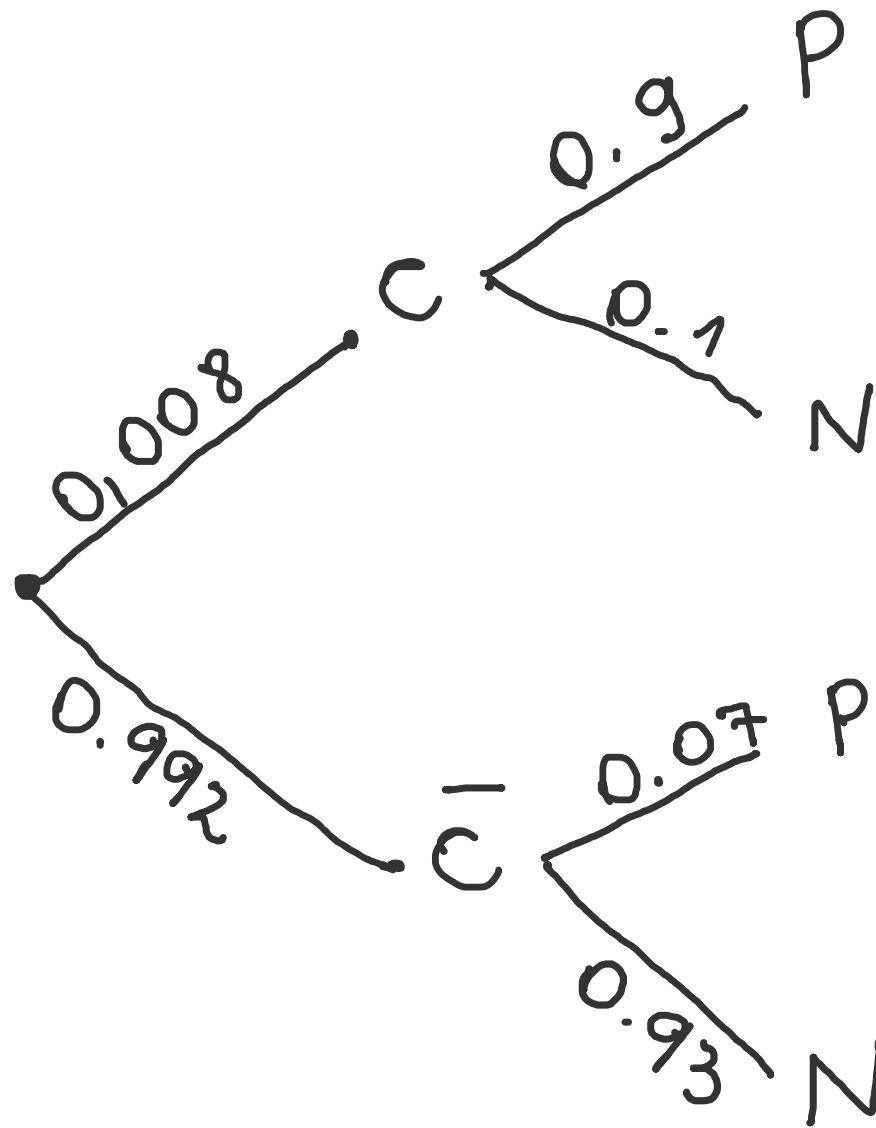
- ▶ What we want to know: *chain rule*

$$\mathbb{P}(C \mid P) = \frac{\mathbb{P}(C \cap P)}{\mathbb{P}(P)} = \frac{\mathbb{P}_r(P \mid C) \cdot \mathbb{P}(C)}{\mathbb{P}_r(P)} = \frac{\mathbb{P}_r(P \mid C) \cdot \mathbb{P}(C)}{\mathbb{P}_r(P \mid C) \cdot \mathbb{P}(C) + \mathbb{P}_r(P \mid \bar{C}) \cdot \mathbb{P}(\bar{C})}$$

law of total probability

MEDICAL TESTING

- ▶ We can use a tree diagram to find $\mathbb{P}(C \cap P)$ and $\mathbb{P}(P)$



$$Pr(C \cap P) = 0.008 \times 0.9$$

$$Pr(P) = 0.008 \times 0.9 + 0.992 \times 0.07$$

$$Pr(C|P) = \frac{Pr(C \cap P)}{Pr(P)}$$

MEDICAL TESTING

- ▶ Another approach via chain rule and law of total probability:
- ▶ $\mathbb{P}(C \cap P) = \mathbb{P}(P \mid C) \cdot \mathbb{P}(C) = 0.9 \cdot 0.008 = 0.0072$
- ▶
$$\mathbb{P}(P) = \underbrace{\mathbb{P}(P \mid C)}_{0.9} \cdot \underbrace{\mathbb{P}(C)}_{0.008} + \underbrace{\mathbb{P}(P \mid \bar{C})}_{0.07} \cdot \underbrace{\mathbb{P}(\bar{C})}_{0.992} = 0.07664$$
- ▶ $\mathbb{P}(C \mid P) = \frac{\mathbb{P}(C \cap P)}{\mathbb{P}(P)} \approx 0.094$

BAYES' THEOREM

- ▶ In our medical testing example, we observed that

$$\mathbb{P}(C \mid P) = \frac{\mathbb{P}(P \mid C)\mathbb{P}(C)}{\mathbb{P}(P)}$$

BAYES' THEOREM

- ▶ In our medical testing example, we observed that


$$\mathbb{P}(C \mid P) = \frac{\mathbb{P}(P \mid C)\mathbb{P}(C)}{\mathbb{P}(P)}$$

let's you swap the events

- ▶ This is called Bayes' theorem and it is a very important result in statistics

$$\mathbb{P}(H \mid X) = \frac{\mathbb{P}(X \mid H) \cdot \mathbb{P}(H)}{\mathbb{P}(X)}$$

H: hypothesis

X: observations

ASK THE AUDIENCE: DOCTORS HAVE A HARD JOB

- ▶ There is a very rare disease affecting 1 in 100,000 people, i.e. a random person has it with probability 0.001%.
- ▶ A random person is selected to take a test with 99% accuracy: a false negative rate of 1% and a false positive rate of 1%.

$\mathbb{P}(\text{have the disease} \mid \text{test is positive})$ is

(A) 99%

(B) About 1 in 10

(C) About 1 in 1,000

(D) About 1 in 10,000

ASK THE AUDIENCE: DOCTORS HAVE A HARD JOB

- There is a very rare disease affecting 1 in 100,000 people, i.e. a random person has it with probability 0.001%.
- A random person is selected to take a test with 99% accuracy: a false negative rate of 1% and a false positive rate of 1%.

$$\Pr(\bar{P}|D) = 0.01$$

$$\Pr(P|\bar{D}) = 0.99$$

$$\Pr(P|\bar{D}) = 0.01$$

$$\Pr(\bar{P}|D) = 0.99$$

$\mathbb{P}(\text{have the disease} \mid \text{test is positive})$ is

$$\Pr(D|P) = \frac{\Pr(P|D) \cdot \Pr(D)}{\Pr(P|D) \cdot \Pr(D) + \Pr(P|\bar{D}) \cdot \Pr(\bar{D})} \approx \frac{0.99 \cdot \frac{1}{10000}}{0.99 \cdot \frac{1}{10000} + \frac{1}{100} \cdot 0.9999} \approx \frac{1}{100}$$

ASK THE AUDIENCE: DOCTORS HAVE A HARD JOB (PART 2)

- ▶ There is a very rare disease affecting 1 in 100,000 people, i.e. a random person has it with probability 0.001%.
- ▶ If you have the disease, you have symptoms. If you have symptoms, the probability you have the disease is 1 in 100. $P(D|S) = 0.99$
- ▶ A random person *with symptoms* is selected to take a test. The test has 99% accuracy: a false negative rate of 1% and a false positive rate of 1%. This remains true for people with or without symptoms.

$\mathbb{P}(\text{have the disease} \mid \text{symptoms and test is positive})$ is

(A) 99% (B) About 1 in 10 (C) About 1 in 1,000 (D) About 1 in 10,000

DOCTORS HAVE A HARD JOB (PART 2)

- ▶ $\mathbb{P}[D] = 0.0001$. $\mathbb{P}[D | S] = 0.01$. $\mathbb{P}[S | D] = 1$.
- ▶ $\mathbb{P}[P | S, D] = 0.99$, $\mathbb{P}[P | S, \bar{D}] = 0.01$.

$$P(P | S \cap D)$$

$$P(P | S \cap \bar{D})$$

$$\mathbb{P}(D | S, P) = \frac{P(P | D \cap S) \cdot P(D | S)}{P(P | D \cap S) \cdot P(D | S) + P(P | \bar{D} \cap S) \cdot P(\bar{D} | S)}$$

$$= \frac{0.99 \cdot 0.01}{0.99 \cdot 0.01 + 0.01 \cdot 0.99}$$

$$= 1/2$$

SUMMARY

Definition of conditional probability: $\Pr(A|B) = \frac{\Pr(A \cap B)}{\Pr(B)}$



chain rule



law of total probability



Baye's rule

$$\Pr(A|B) = \frac{\Pr(B|A) \cdot \Pr(A)}{\Pr(B)}$$

$$\Pr(A) = \sum_{i=1}^K \Pr(A|B_i) \cdot \Pr(B_i)$$

where $B_1, \dots, B_K$ are partitions of the sample space Ω